

Lab 1-1

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1. Bubble sort works by iterating over the list multiple times. During each iteration, it compares each pair of ~~adj~~ adjacent elements and swaps them if they are in the wrong order. This way, the largest element moves to ~~the~~ its correct ~~pos~~ position at the end of the list after each iteration. The process is repeated until no swaps are needed.
2. Insertion sort works by iteratively inserting each element of an unsorted list into its correct position in a sorted portion of ~~the~~ the list. Splitting the list into two groups: the sorted list and the unsorted list. Then pick an element from unsorted list and put it in the right place in the sorted list.
3. Merge sort uses divide and conquer idea. Firstly, dividing the list recursively into two halves until it can no more be divided. Secondly, each subarray is sorted ~~individually~~ individually using the merge sort algorithm. Finally, the sorted subarrays are merged back together in sorted order.